# CUSTOMER SHOPPING BEHAVIOR ANALYSIS

*Data Analytics Project Report*

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3,900 Customers | 4 Product Categories | Python + SQL + Power BI

## 1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover actionable insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

The analysis follows an end-to-end data pipeline:

- **Data Cleaning & Preprocessing** using Python (pandas, numpy)
- **Exploratory Data Analysis** with statistical summaries and feature engineering
- **Business Intelligence Querying** using SQL (MySQL/PostgreSQL)
- **Interactive Dashboard** built in Power BI for stakeholder reporting

## 2. Dataset Summary

The dataset captures retail transactional data with the following key characteristics:

| Attribute | Details |
|---|---|
| Total Records | 3,900 |
| Total Columns | 18 |
| Missing Values | 37 (Review Rating column) |
| Average Purchase Amount | $59.76 |
| Average Review Rating | 3.75 / 5.0 |
| Date Coverage | Multi-season (Spring, Summer, Fall, Winter) |

Key feature groups in the dataset:

- **Customer Demographics:** Age, Gender, Location, Subscription Status
- **Purchase Details:** Item Purchased, Category, Purchase Amount, Season, Size, Color
- **Shopping Behavior:** Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type

# 3. Exploratory Data Analysis (Python)

Data preparation and cleaning was performed using Python with the pandas library. The following steps were taken in sequence:

## 3.1 Data Loading & Initial Exploration

The dataset was imported using pandas. Initial exploration used df.info() to inspect column types and non-null counts, and df.describe(include='all') for descriptive statistics across both numerical and categorical features.

## 3.2 Missing Data Handling

The Review Rating column contained 37 missing values. These were imputed using the median rating grouped by product Category — ensuring no cross-category bias was introduced:

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x:
x.fillna(x.median()))
```

## 3.3 Column Standardization

All column names were converted to snake_case for consistency and ease of use in SQL. The purchase_amount_(usd) column was renamed to purchase_amount for clarity.

## 3.4 Feature Engineering

- **age_group:** Customer ages were binned into four quartile-based groups: Young Adult, Adult, Middle Aged, and Senior.
- **purchase_frequency_days:** The categorical frequency_of_purchases column was mapped to numeric day values (e.g., Weekly → 7, Monthly → 30, Annually → 365).

## 3.5 Data Consistency & Redundancy Check

A verification check confirmed that the discount_applied and promo_code_used columns contained identical values. The promo_code_used column was therefore dropped to eliminate redundancy.

## 3.6 Database Integration

The cleaned DataFrame was exported to a MySQL database (customer_shopping schema) using SQLAlchemy and PyMySQL, enabling efficient SQL-based analysis in subsequent steps.

# 4. Data Analysis Using SQL

Structured business queries were executed in MySQL/PostgreSQL to answer ten key business questions. Results are presented below:

## Q1: Revenue by Gender

Total revenue was compared between male and female customers. Male customers contributed significantly more to overall revenue.

| Gender | Total Revenue (USD) |
|---|---|
| Female | $75,191 |
| Male | $157,890 |

Insight: Male customers account for approximately 68% of total revenue. Targeted campaigns toward female customers could help balance revenue distribution.

## Q2: High-Spending Discount Users

839 customers used discounts yet still spent above the average purchase amount of $59.76. This indicates that discount users are not necessarily low-value customers — many are high-intent buyers who respond well to promotions.

| Metric | Value |
|---|---|
| Customers with Discount & Above-Avg Spend | 839 |
| Average Purchase Amount (All) | $59.76 |
| Minimum Qualifying Amount | $60+ |

## Q3: Top 5 Products by Average Review Rating

Products were ranked by average customer review rating. Gloves topped the list, followed by Sandals and Boots.

| Rank | Product | Avg. Review Rating |
|---|---|---|
| 1 | Gloves | 3.86 |
| 2 | Sandals | 3.84 |
| 3 | Boots | 3.82 |
| 4 | Hat | 3.80 |
| 5 | Skirt | 3.78 |

## Q4: Shipping Type Comparison

Average purchase amounts were compared between Standard and Express shipping customers.

| Shipping Type | Avg. Purchase Amount (USD) |
|---|---|
| Standard | $58.46 |
| Express | $60.48 |

Insight: Express shipping customers spend slightly more on average, suggesting they may be higher-value customers who prioritize speed — a potential segment for premium upselling.

## Q5: Subscribers vs. Non-Subscribers

Subscription status was analyzed against total customers, average spend, and total revenue.

| Subscription Status | Total Customers | Avg. Spend (USD) | Total Revenue (USD) |
|---|---|---|---|
| Yes | 1,053 | $59.49 | $62,645 |
| No | 2,847 | $59.87 | $170,436 |

Insight: Despite making up only 27% of customers, subscribers contribute $62,645 in revenue. Non-subscribers have a marginally higher average spend, suggesting an opportunity to convert high-spenders into subscribers through exclusive perks.

## Q6: Discount-Dependent Products

The top 5 products with the highest percentage of discounted purchases were identified. These products may be overly reliant on discounts to drive sales.

| Rank | Product | Discount Rate (%) |
|---|---|---|
| 1 | Hat | 50.00% |
| 2 | Sneakers | 49.66% |
| 3 | Coat | 49.07% |
| 4 | Sweater | 48.17% |
| 5 | Pants | 47.37% |

Insight: Approximately half of Hat and Sneaker purchases involve discounts. Consider testing price optimization or bundling strategies for these items to reduce discount dependency.

## Q7: Customer Segmentation

Customers were segmented into three groups based on number of previous purchases: New (1), Returning (2–10), and Loyal (>10).

| Customer Segment | Number of Customers |
|---|---|
| Loyal | 3,116 |
| Returning | 701 |
| New | 83 |

Insight: An impressive 80% of the customer base is already in the Loyal segment, indicating strong retention. Focus should be on converting New and Returning customers into Loyal ones through targeted loyalty programs.

## Q8: Top 3 Products per Category

The most frequently purchased items were identified within each product category using window functions.

| Category | Rank 1 | Rank 2 | Rank 3 |
|---|---|---|---|
| Accessories | Jewelry (171) | Sunglasses (161) | Belt (161) |
| Clothing | Blouse (171) | Pants (171) | Shirt (169) |
| Footwear | Sandals (160) | Shoes (150) | Sneakers (145) |
| Outerwear | Jacket (163) | Coat (161) | Hoodie (~145) |

## Q9: Repeat Buyers & Subscription Likelihood

Among customers with more than 5 previous purchases, subscription adoption was examined.

| Subscription Status | Repeat Buyers (>5 Purchases) |
|---|---|
| No | 2,518 |
| Yes | 958 |

Insight: Even among the most active repeat buyers, subscription adoption is low (28%). This is a major growth opportunity — a loyalty-to-subscription conversion funnel could significantly boost recurring revenue.

## Q10: Revenue by Age Group

Total revenue contribution was calculated for each age group, revealing that Young Adults are the highest-spending demographic.

| Age Group | Total Revenue (USD) |
|---|---|
| Young Adult | $62,143 |

| Age Group | Total Revenue (USD) |
|---|---|
| Middle-Aged | $59,197 |
| Adult | $55,978 |
| Senior | $55,763 |

Insight: Revenue is fairly balanced across age groups, with Young Adults leading slightly. All segments are valuable, but youth-oriented marketing could yield the highest returns.

# 5. Power BI Dashboard

An interactive dashboard was built in Power BI to present findings visually for stakeholders. The dashboard includes dynamic filters for Subscription Status, Gender, Category, and Shipping Type, allowing users to slice data interactively.

Key visualizations included in the dashboard:

- KPI Cards: Total Customers (3,900), Average Purchase Amount ($59.76), Average Review Rating (3.75)
- Donut Chart: Subscription Status breakdown (27% subscribers vs 73% non-subscribers)
- Bar Charts: Revenue and Sales by Category (Clothing leads both)
- Horizontal Bar Charts: Revenue and Sales by Age Group

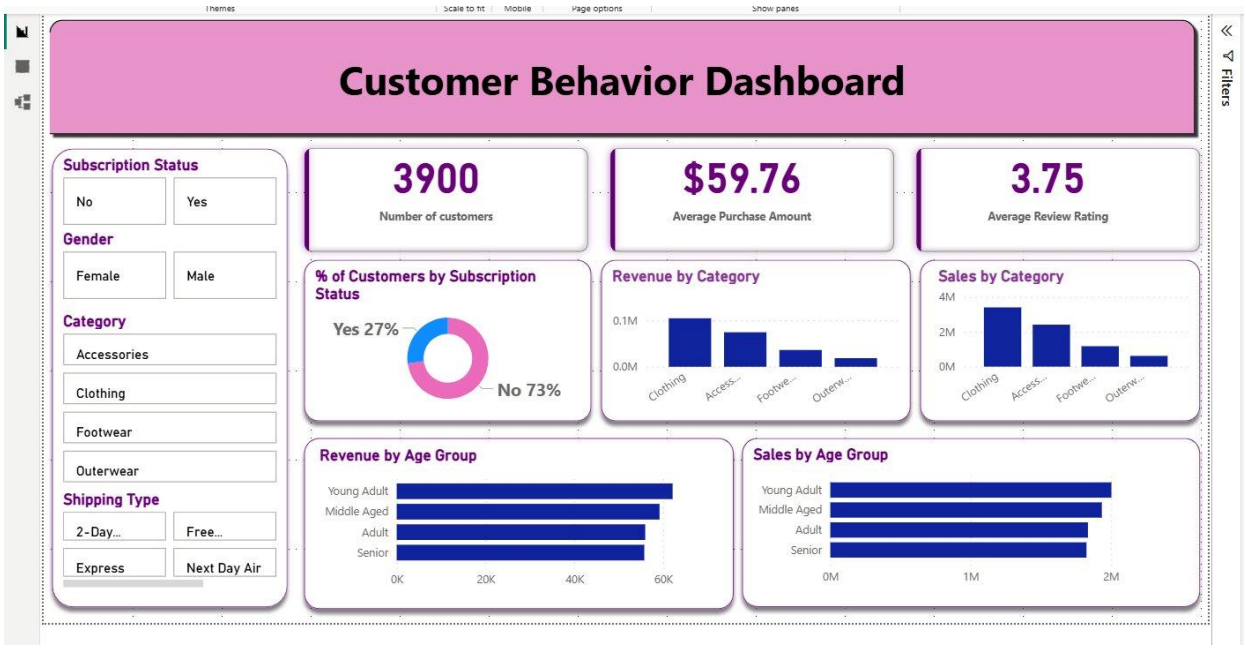


*Figure 1: Customer Behavior Dashboard — Power BI*

# 6. Business Recommendations

Based on the analysis, the following strategic recommendations are proposed:

- **Boost Subscription Conversions:** Only 27% of customers subscribe. Given that repeat buyers with >5 purchases still have low subscription rates, a targeted funnel offering exclusive discounts, early access, or free shipping for subscribers could drive conversion significantly.

- **Invest in Customer Loyalty Programs:** 80% of customers are already Loyal. Formalize this with a tiered rewards program to deepen engagement, increase average order value, and create brand advocates.

- **Review Discount Strategy:** Products like Hats and Sneakers have ~50% discount rates. Conduct margin analysis to determine whether these discounts are profitable or eroding margins. Consider limited-time or bundle-based offers instead.

- **Capitalize on Express Shipping Customers:** Express shipping users spend slightly more on average. Target this segment with premium product recommendations and subscription upsells.

- **Product Positioning & Campaigns:** Gloves, Sandals, and Boots have the highest review ratings. Feature these in marketing materials and homepage placements to leverage social proof.

- **Targeted Marketing by Demographics:** Young Adults generate the highest revenue. Tailor campaigns with trends, social media, and influencer partnerships aimed at this cohort. Simultaneously, create senior-friendly UX and product recommendations to grow that segment.

- **Gender Revenue Gap Analysis:** Male customers generate 2.1x more revenue than female customers. Investigate whether this reflects product-mix bias or a marketing gap, and develop female-targeted campaigns or product lines to close the gap.

# 7. Tools & Technologies

| Tool / Technology | Purpose |
|---|---|
| Python (pandas, numpy) | Data loading, cleaning, feature engineering |
| SQLAlchemy + PyMySQL | Python-to-MySQL database connection |
| MySQL / PostgreSQL | SQL-based business intelligence queries |
| Power BI | Interactive dashboard and data visualization |
| Jupyter Notebook | Development environment for Python EDA |

# 8. Conclusion

This end-to-end data analytics project successfully transformed raw transactional data into actionable business intelligence. By combining Python's data manipulation capabilities, SQL's querying power, and Power BI's visualization strengths, we uncovered key patterns in customer behavior — from spending habits and product preferences to the impact of discounts and subscription status.

The most significant findings include the strong loyalty of the existing customer base, the gender revenue gap, the untapped subscription growth potential, and the discount dependency of certain high-volume products. These insights provide a clear roadmap for data-driven decision-making across marketing, product, and customer success teams.

*End of Report*